

Report-Social Media Effects on Mental Health

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Abstract

This project investigates the complex relationship between social media engagement and mental health, specifically exploring whether high screen time and negative online interactions correlate with high-risk mental states, poor sleep, stress, and anxiety. The study utilizes a dataset from the years 2024-2025 containing 5,000 questionnaire responses, resulting in 3,949 clean user profiles across 15 distinct behavioral and psychological features. To preserve the absolute integrity of the genuine statistical patterns, the analytical pipeline maintained the original clean data rather than injecting artificial noise.

Our primary project question explores whether the choice of the primary social media platform, alongside user age, correlates with user anxiety or stress levels.

The core methodology relies on Non-Negative Matrix Factorization (NMF) to compress the 15-dimensional space down to 3 additive components, successfully extracting interpretable behavioral archetypes rather than abstract mathematical axes. A Gaussian Mixture Model (GMM) was implemented as the primary clustering mechanism to allow for soft probabilistic assignments, accurately modeling the nuanced spectrum of human behavior.

The primary findings revealed three distinct behavioral segments. The "Heavy Social Media User" cluster is characterized by extreme TikTok and Instagram usage, high online volatility, and severe anxiety. This is contrasted against the "Utility Screen User," who primarily uses WhatsApp and maintains moderate strain, and the older "Healthy, Low-Digital User," who favors Facebook, maintains high physical activity, and experiences the lowest mental strain. Ultimately, the analysis proves that platform design, specifically visual-heavy algorithms versus communication-focused utilities, dictates mental health outcomes just as much as raw screen time.

Introduction

The application domain of this study focuses on the intersection of public mental health, behavioral psychology, and digital consumer habits. In an era where digital connectivity is ubiquitous, motivating this project is the urgent need to understand the hidden psychological tax of modern screen habits. With growing societal concern over algorithmic amplification and its impact on user well-being, mapping out how online interactions dictate daily biological rhythms is crucial.

The dataset is highly interesting because it explicitly separates general screen time from dedicated social media time, pairing these habits directly with clinical self assessments of anxiety, stress, and mood levels, alongside lifestyle variables like sleep hours and physical activity. This creates a comprehensive multidimensional profile for every user. The central hypothesis being investigated is that the choice of a primary social media platform and the intensity of screen engagement directly correlate with high risk mental states and severe sleep disruptions.

Traditional statistical methods evaluate variables in isolation (for example, testing only how screen time affects sleep), which completely fails to capture how a person's entire lifestyle interacts. Clustering is uniquely powerful here because it acts as a multidimensional lens that analyzes all 15 behavioral and psychological features simultaneously. Instead of forcing users into arbitrary, pre-defined categories, clustering allows us to discover natural, data-driven population profiles based on global behavioral footprints.

By grouping individuals who share similar combinations of platform choices, sleep patterns, and stress levels, clustering reveals the hidden behavioral archetypes within society. This directly helps answer our project questions by showing us exactly which holistic lifestyles map to high-risk mental states versus healthy ones. Furthermore, because human psychology and digital habits do not have rigid boundaries, using a probabilistic clustering approach like GMM provides the necessary flexibility to model these highly correlated features and soft overlapping tendencies accurately.

Dataset Description

what each sample represents

questionnaire completed by an individual in a certain date, detailing their phone and social media usage, as well as their physical and mental health status

what the features represent

person_name - The name of the user

age - The age of the user

date - The date on which the data was recorded

gender - Gender of the user (Male, Female, Other)

platform - Primary social media platform the person uses (Facebook, Instagram, Snapchat, TikTok, Twitter, Whatsapp, Youtube)

daily_screen_time_min - Total daily device screen time in minutes

social_media_time_min - Total time spent on social media in minutes per day

negative_interactions_count - number of negative or harmful interactions experienced online

positive_interactions_count - number of positive interactions experienced online

sleep_hours - Total number of hours the person sleeps per day

physical_activity_min - Daily physical activity in minutes

anxiety_level - Self-reported anxiety score

stress_level - Self-reported stress score

mood_level - Self-reported mood score

mental_state - The examiner labeled mental state (Healthy, At_Risk, Stressed)

the size of the dataset

number of samples (rows) - 5000

number of features (columns) - 15

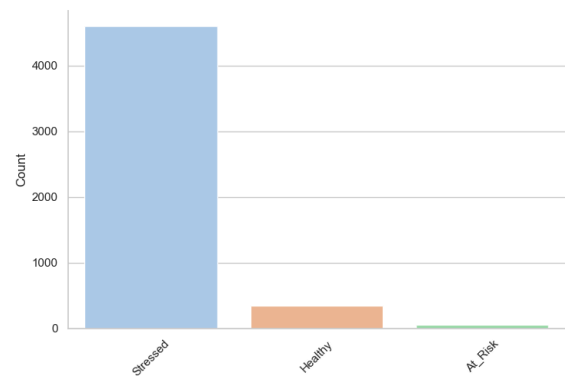
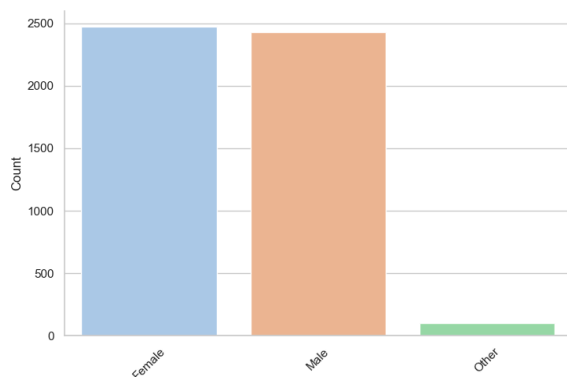
any limitations of the data

Extreme Data Imbalance (Class Imbalance): the dataset exhibits a severe imbalance in the target variable's distribution. Specifically, 92.02% of the users are classified as 'Stressed', while only 6.82% are 'Healthy' and a mere 1.16% are 'At_Risk'. Clustering algorithms naturally tend to bias their learning toward the dominant majority class. Consequently, the models may struggle to accurately isolate and capture the distinct behavioral patterns of the underrepresented minority groups.

Exploratory Data Analysis

feature distributions

The numerical features possess drastically different natural magnitudes. For instance, time-based metrics (like daily screen time) span hundreds of minutes, whereas self-reported psychological scores (anxiety, stress, mood) are restricted to a narrow 1 to 10 scale. Furthermore, there is a severe class imbalance in the target demographic: 92.02% of the users are classified as Stressed, while only 6.82% are Healthy and 1.16% are At_Risk. In contrast, the gender distribution is nearly symmetrical (49.48% Female, 48.54% Male).



missing values

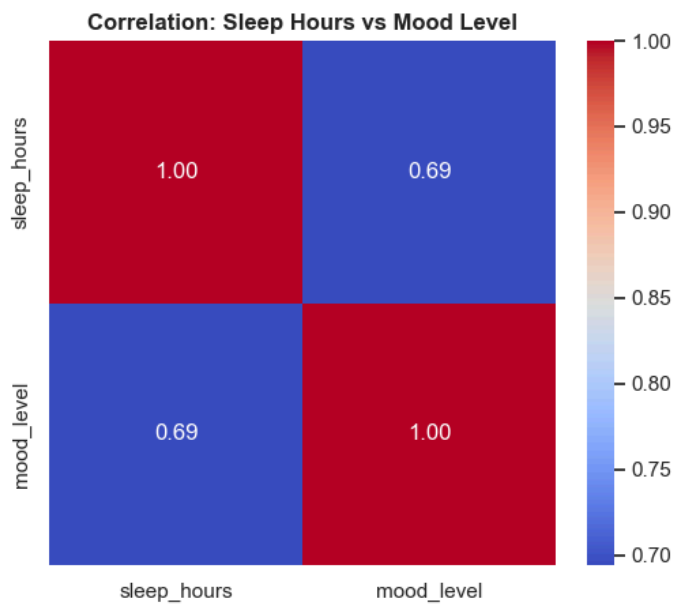
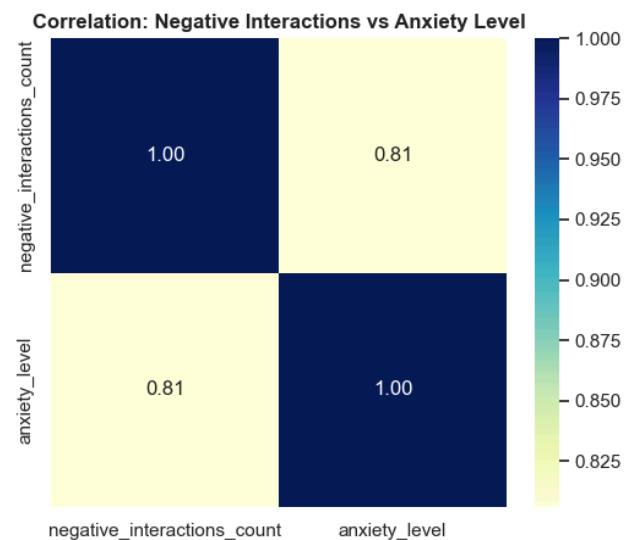
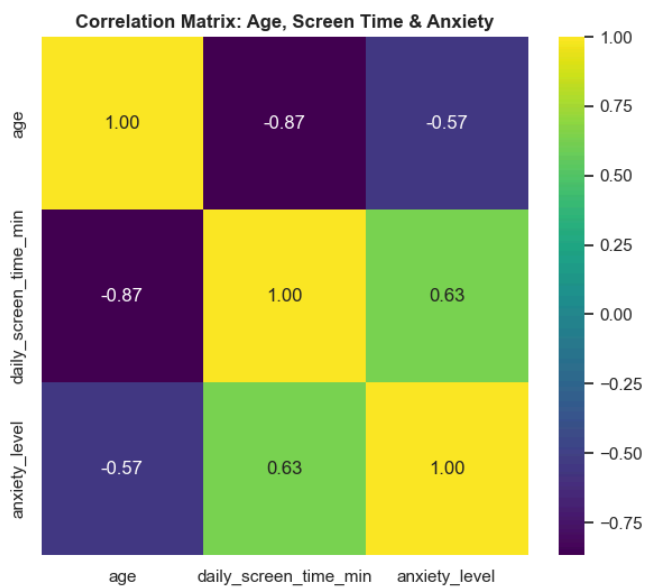
The dataset was inherently clean, containing zero missing values, nulls, or typos. We researched various methods to intentionally "dirty" the data (such as injecting artificial noise or missing values to simulate real-world messiness). We decided to continue working with our clean, original dataset to preserve absolute data integrity and ensure authentic clustering results.

outliers

Following the same reasoning, we chose not to inject extreme artificial outliers.

interesting patterns discovered before clustering

The correlation heatmaps exposed several strong relationships within the data. Most notably, there is a strong negative correlation (-0.87) between Age and Daily Screen Time, indicating that younger users spend significantly more time on screens. Additionally, Negative Interactions and Anxiety Level show a strong positive correlation (0.81), while Sleep Hours positively correlate with Mood Level (0.69). There is also a moderate positive correlation (0.63) between Daily Screen Time and Anxiety Level.



Preprocessing

Encoding Categorical Features

The dataset contained both nominal and ordinal categorical features that required transformation into a machine-readable format. We applied ordinal encoding to the mental state variable (Healthy = 0, Stressed = 1, At Risk = 2) to preserve its inherent severity hierarchy. For nominal features without an inherent order, such as gender and platform, we applied One-Hot Encoding to prevent the models from inferring false mathematical relationships.

Scaling and Normalization

The numerical features exhibited drastically different natural scales. For instance, time-based metrics span hundreds of minutes, whereas psychological scores are restricted to a 1 to 10 scale. To ensure that large-magnitude variables do not disproportionately dominate distance calculations in our clustering algorithms, we applied Min-Max Scaling. This successfully standardized all numerical variables to a uniform range between 0 and 1.

Feature Engineering

We engineered several new features to extract deeper relative insights and better organize the data structure.

First, we created composite ratio columns, such as social media ratio and activity to screen ratio, to capture relative behavioral proportions rather than relying solely on absolute minutes.

Second, we generated a unique identifier for each user to resolve ambiguities. Recognizing that multiple records shared the same person name but belonged to different individuals, we grouped records by identical names and applied a one-year age buffer. This method allowed us to accurately assign a precise unique ID to each person. Through this process, we discovered that the 5000 samples in our dataset actually originate from exactly 3949 distinct individuals.

Representation

Dimensionality reduction is highly beneficial for this dataset. It mitigates noise, handles multicollinearity between correlated psychological metrics, and distills the features into core behavioral patterns before passing them into clustering algorithms.

Method Selected and Justification

We selected **Non-Negative Matrix Factorization (NMF)** for this analysis. Unlike Principal Component Analysis (PCA), which allows for negative loadings and orthogonal vectors, NMF enforces strictly positive additive weights. Because our preprocessed dataset relies on strictly non-negative scaled data, NMF organically groups features into cohesive "parts". This additive nature means the resulting components naturally represent distinct, real-world user behavioral profiles rather than abstract mathematical axes.

Selected Parameters

The following parameters were initialized for the NMF model: `n_components=3`, `init='random'`, `random_state=42`, and `max_iter=500`. A rank of 3 was chosen because it aggressively reduces the feature space while successfully capturing the three primary dimensions of variance in the data.

The effect on clustering performance

Applying Gaussian Mixture Models (GMM) on the NMF-reduced space yielded excellent clustering cohesion. When evaluating the optimal clustering for the NMF (3 Components) representation, the algorithm identified **K=3 clusters** utilizing a **Spherical** covariance structure as the optimal fit.

- **NMF1 (The Stressed Social Media User):** Driven entirely by high mental strain and social media consumption. Top influencing weights include `mental_state` (0.83), `social_media_ratio` (0.77), and `anxiety_level` (0.75).
- **NMF2 (The Healthy/Active User):** Driven by positive physical health indicators and better mood. Top influencing weights include `sleep_hours` (0.80), `physical_activity` (0.80), and `mood_level` (0.74).
- **NMF3 (The General Screen User):** Driven by high overall screen time independent of social media platforms. Top influencing weights include `mental_state` (0.91) and `daily_screen_time_w/o_social_media` (0.76).

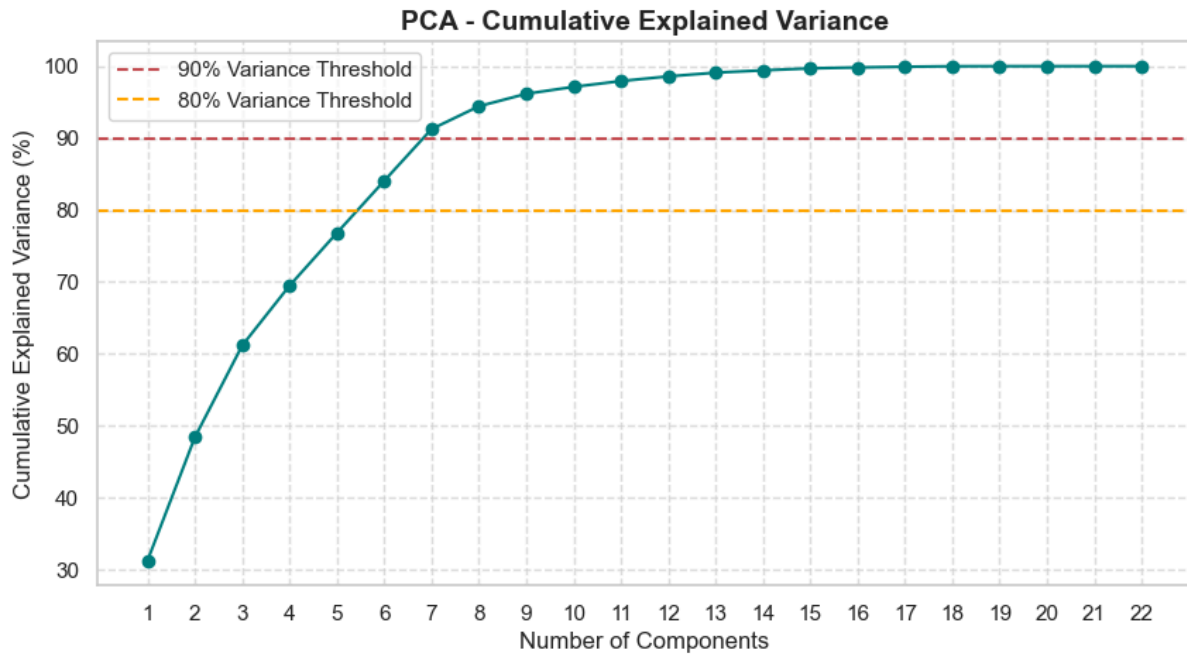
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Advantages and Limitations

- **Advantages:** The primary advantage is the intuitive, parts-based representation of the data. NMF allowed us to feed mathematically distinct personas directly into the GMM algorithm, which subsequently mapped perfectly to 3 highly cohesive clusters. It isolated the psychological distress variables from the healthy habits efficiently.

- **Limitations:** NMF mandates that input data be strictly non-negative, meaning robust preprocessing via MinMaxScaler was a hard requirement. Furthermore, because NMF does not force orthogonal components like PCA does, there is potential for feature overlap and correlation between the resulting latent components.



Clustering Analysis

To effectively segment the user profiles, we evaluated two distinct clustering algorithms, analyzing their performance across the entire 5,000-row dataset.

Algorithms Evaluated and Selected Parameters

We implemented and compared a distance-based approach and a probabilistic approach:

- **Hierarchical Clustering (Agglomerative):** A bottom-up approach that starts with each data point as its own cluster and progressively merges them. We utilized a grid-search to evaluate it across four linkage methods ('ward', 'complete', 'average', and 'single') to determine the optimal distance measurement between clusters .
- **Gaussian Mixture Models (GMM):** A probabilistic model assuming data points are generated from a mixture of finite Gaussian distributions. We evaluated GMM across four covariance structures ('full', 'tied', 'diag', and 'spherical') to determine the most natural shape for the clusters.

Choosing the Number of Clusters (K) A rigorous, multi-step optimization sequence was used to prevent arbitrary cluster selection. First, we applied the **Elbow Method** (using K-Means from K=2 to 10) to calculate the Within-Cluster Sum of Squares (WCSS). To pinpoint the exact "elbow" objectively, we implemented a custom mathematical function to find the point on the WCSS curve with the maximum geometric distance to the secant line connecting the endpoints, establishing a base K. Finally, we used **Silhouette Score Validation** testing the baseline K alongside K+1 and K+2 across all covariance types to definitively lock in the highest scoring and most cohesive configuration .

Advantages and Disadvantages

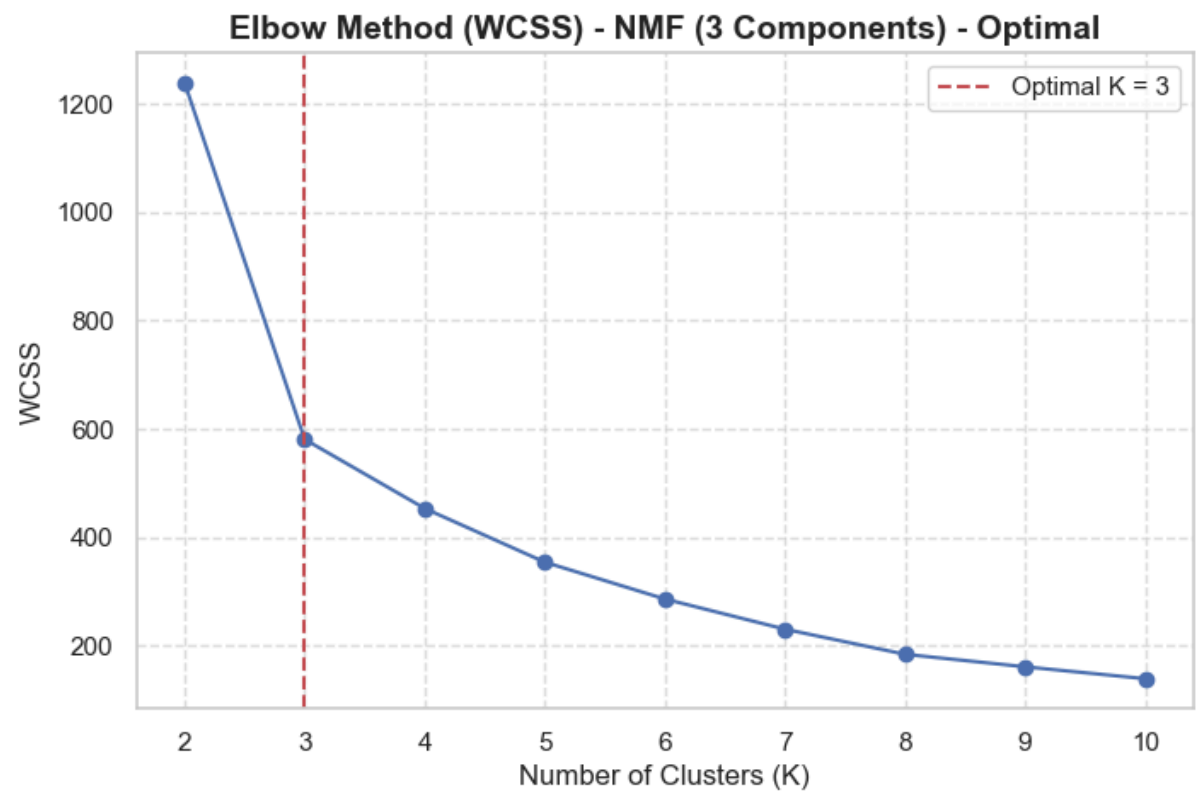
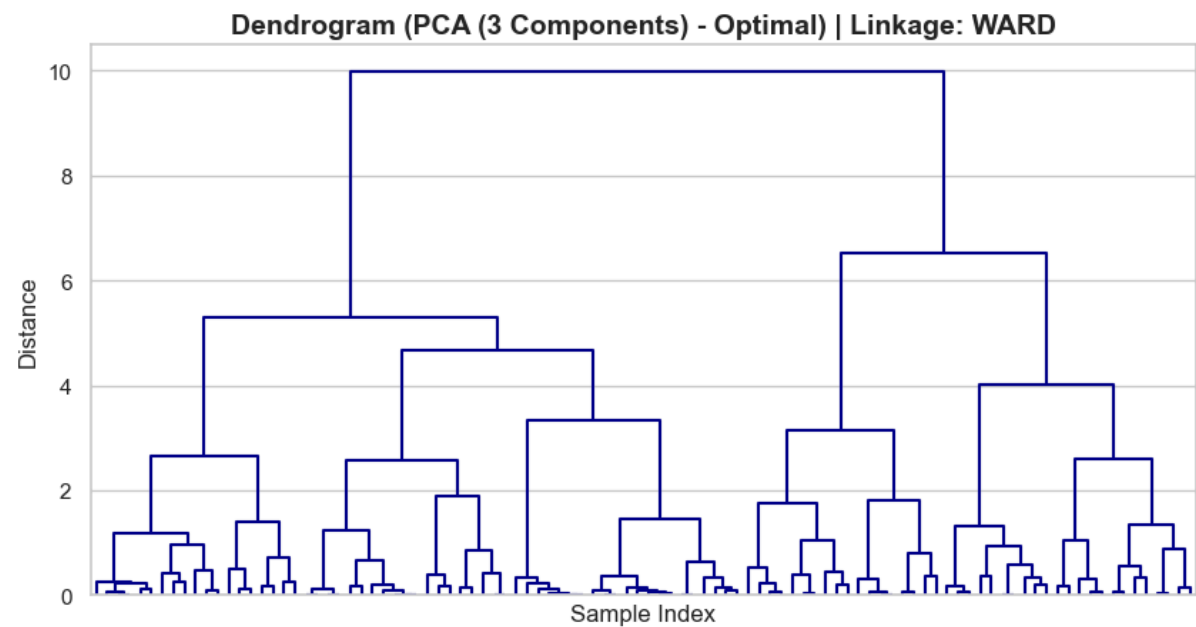
- **Hierarchical Clustering:** The primary advantage is its transparency, it produces highly interpretable dendrograms that visually track the exact hierarchical merges of user profiles. However, it is strictly rigid. Once two users are merged, they cannot be separated later. Additionally, it is computationally expensive for the full 5,000-row dataset, operating at an $\mathcal{O}(N^3)$ time complexity for some linkages.
- **Gaussian Mixture Models (GMM):** GMM is incredibly flexible. By altering covariance types, it adapts to natural data distributions (elliptical or spherical) rather than forcing rigid geometric boundaries. Crucially, it provides "soft assignments," calculating the probability of a user belonging to a cluster. The main disadvantage is the risk of overfitting if given an overly complex covariance structure (like 'full'), though our reliance on Silhouette score optimization effectively mitigated this risk.

Final Selection and Justification

Gaussian Mixture Models (GMM) was ultimately selected as the primary clustering method for this project.

As the pipeline advanced to utilizing Non-Negative Matrix Factorization (NMF), GMM consistently outperformed Hierarchical clustering. When analyzing the complex psychological and behavioral variables mapped by the NMF (The Stressed User, The

Healthy User, The General Screen User), GMM utilizing a **Spherical** covariance structure yielded the highest Silhouette Score at exactly **K=3** clusters. Ultimately, GMM was preferred because human behavior is rarely absolute. The probabilistic "soft boundaries" of GMM perfectly accommodate the overlapping nuances of social media habits and mental health states in a way that rigid, distance-based hierarchical models simply cannot.

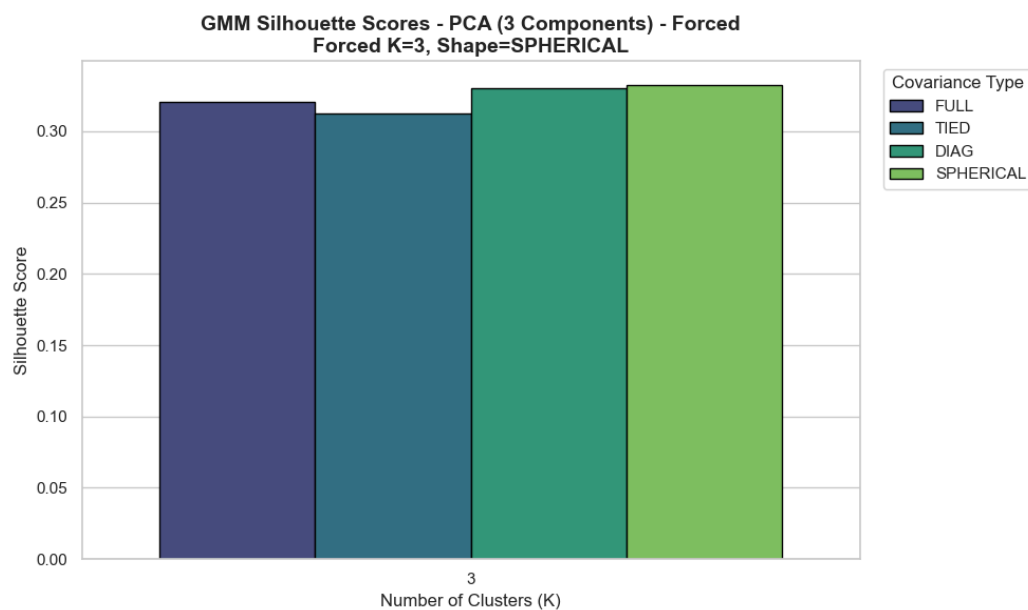
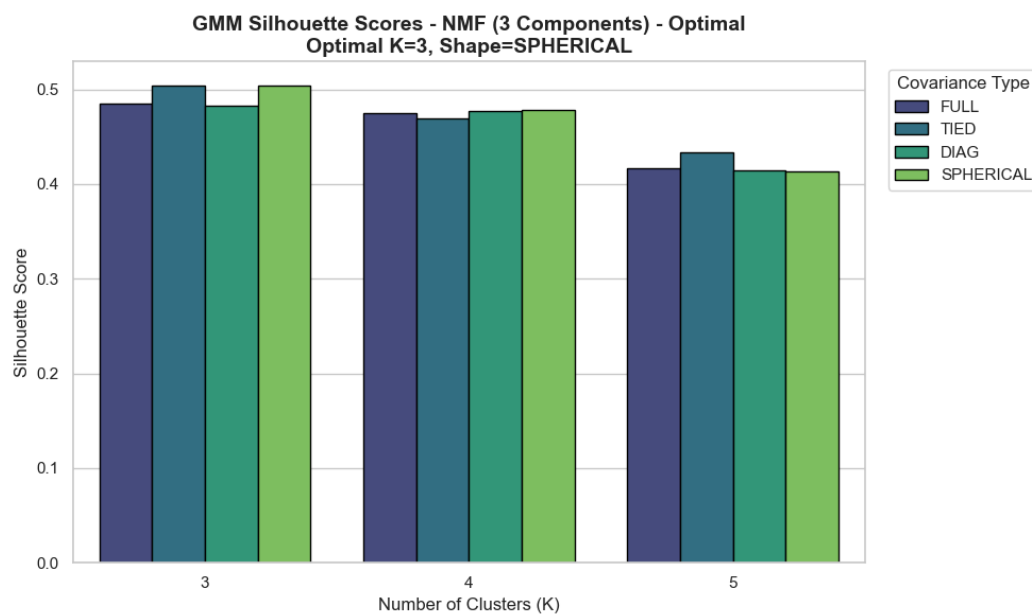


Cluster Evaluation and Visualization

Evaluation Scores (Silhouette Score)

The silhouette score measures how similar an object is to its own cluster compared to other clusters. Across the various pipelines, **GMM consistently matched or outperformed Hierarchical Clustering** in generating cohesive groups:

- **PCA Representations:** Both algorithms struggled to find highly cohesive clusters using PCA. For example, in the PCA (3 Components) pipeline, Hierarchical (Ward linkage) scored ~0.31, while GMM (Spherical, K=4) scored ~0.35.
- **NMF Representation:** When using Non-Negative Matrix Factorization (3 Components), GMM with a Spherical covariance structure rebounded to a strong silhouette score of approximately 0.5 at K=3.

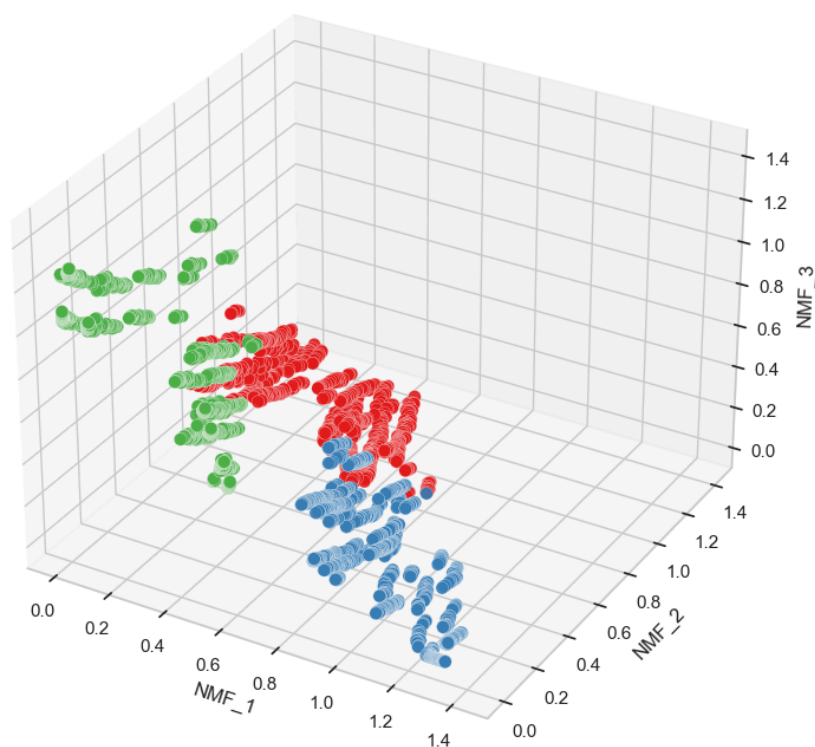


Visual inspection of the resulting clusters

Both algorithms generally identified similar macro-patterns separating users with heavy screen time and high stress from those with healthier digital habits. However, visual inspection reveals stark differences in how they draw the boundaries between these groups:

- **Hierarchical Clustering:** Visualized through the dendrograms, this method forms rigid, definitive splits between user profiles. While mathematically sound, the 2D and 3D scatter plots show that these rigid boundaries sometimes arbitrarily slice through dense, continuous clouds of user data.
- **GMM:** The 3D scatter plots for GMM demonstrate its superior ability to map the actual shape of the data. Because GMM utilizes covariance structures (like Spherical or Diag), it clusters the 5,000 data points into naturally occurring, overlapping spheres or ellipses, which visually aligns much better with the dense data distributions.

GMM 3D Clustering (Optimal K=3, Shape=spherical) - NMF (3 Components) - Optimal



Comparison and Meaningfulness

While the Manual Dimensionality Reduction achieved the highest mathematical silhouette scores, GMM paired with NMF (3 Components) produced the most meaningful and interpretable clusters.

The profiling dashboards generated by the NMF + GMM (K=3) pipeline cleanly segment the users into highly distinct, real-world behavioral profiles.

- **Segment 1:** captures users with the highest daily screen time and lowest physical activity.
- **Segment 2:** clearly isolates the users experiencing the highest psychological strain (Anxiety and Stress levels).
- **Segment 3:** identifies healthier, more active users.

Hierarchical clustering, by contrast, forces a "hard" assignment. Because mental health and social media habits exist on a spectrum. For example, a user can be both active and highly stressed. The probabilistic "soft assignments" of GMM inherently **reflect the nuanced reality of human behavior much better.**

Final Recommendation

We recommend utilizing **Gaussian Mixture Models (GMM)** paired with **Non-Negative Matrix Factorization (NMF at 3 components)**, optimized for **K=3 clusters**.

GMM accommodates the overlapping, continuous nature of the 5,000-row dataset through probabilistic clustering and flexible covariance structures, whereas Hierarchical clustering is too rigid. Furthermore, while manual reduction scored higher mathematically, it risks introducing human bias. NMF organically distills the dataset into three highly interpretable parts (Stress, Health, General Use). When GMM clusters these NMF components, it yields a strong silhouette score (~0.5) while producing the most actionable and logically coherent user personas for analysis.

Cluster Interpretation

Cluster 1: The "Utility Screen" User (Moderate Strain)

- **Main Characteristics:** This group represents a young demographic (average age ~25) with low physical activity (~18 minutes/day) and average sleep (~7 hours) . They spend a massive amount of time on screens (~380 minutes/day), but a relatively low portion of that is spent on social media (~140 minutes/day) . Their psychological scores sit in the middle: moderate stress (~6.1), moderate anxiety (~2.1), and an average mood (~5.2) . They heavily favor communication/information platforms like WhatsApp and Twitter .
- **Distinguishing Features:** They have the largest gap between total daily screen time and social media time, meaning the bulk of their screen usage (~280 minutes) is non-social . Their online interaction counts (both positive and negative) are relatively low compared to other heavy screen users .
- **Application Meaningfulness:** Highly meaningful. This cluster effectively captures the modern "digital worker" or "utility scroller." Their high screen time is likely tied to daily life, work, or non-social entertainment. Because they are not deeply tethered to algorithmically driven social feeds, they experience baseline modern stress but avoid the extreme anxiety and volatility seen in heavy social media users.

Cluster 2: The "Heavy Social Media" User (High Psychological Strain)

- **Main Characteristics:** Also a young demographic (average age ~24) with low physical activity (~18 minutes) and average sleep (~7 hours) . They exhibit the highest overall screen time (~390 minutes/day), with the majority dedicated directly to social media (~240 minutes/day) . They suffer from the worst psychological outcomes: peak Stress (~8.0), peak Anxiety (~3.1), and the lowest Mood (~5.1) . They favor highly engaging, short-form visual platforms like TikTok, Snapchat, and Instagram .
- **Distinguishing Features:** This segment is defined by extreme online volatility. They log massively higher counts of both positive interactions (~27) and negative interactions (~12) than any other group . Furthermore, nearly 100% of the users in this cluster are categorized under a "Stressed" mental state .
- **Application Meaningfulness:** Extremely meaningful. This cluster perfectly encapsulates the core hypothesis of digital wellness research: heavy reliance on highly engaging visual social media correlates directly with severe psychological strain, online volatility, and heightened anxiety.

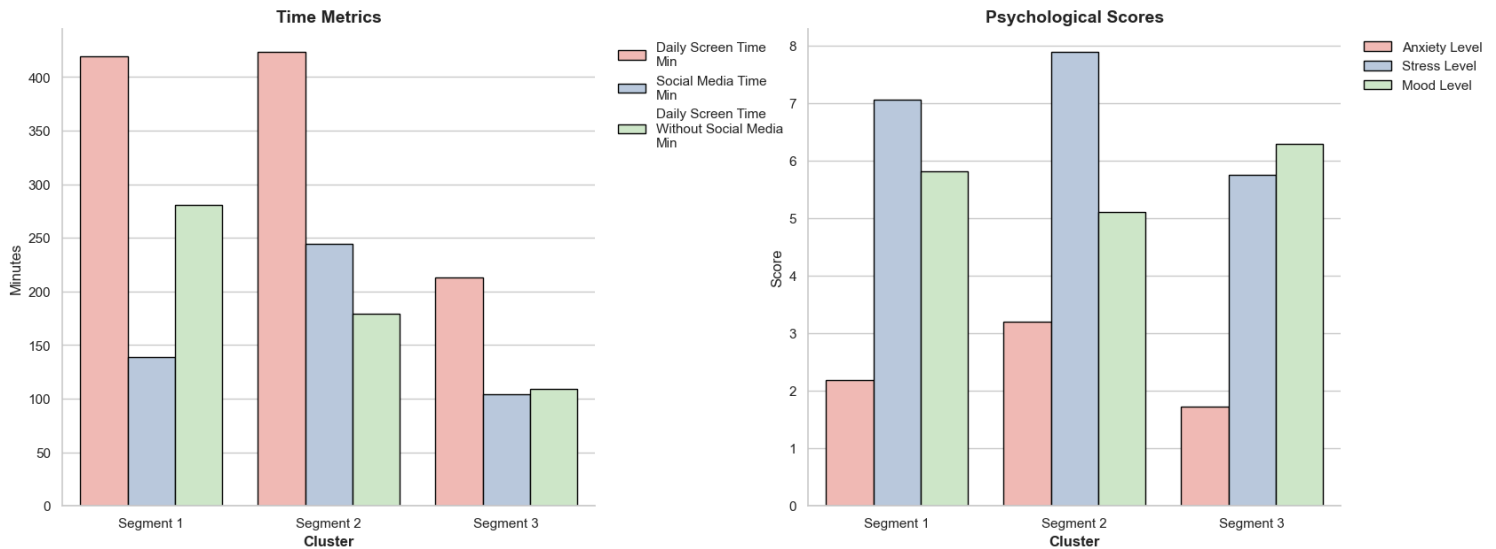
Cluster 3: The "Healthy, Low-Digital" User (Low Strain)

- **Main Characteristics:** A significantly older demographic (average age ~48) . They have the best offline habits: highest physical activity (~39 minutes/day) and the most sleep (~8 hours/day) . Consequently, they have the lowest screen time (~210 minutes/day) and lowest social media time (~100 minutes/day) . They boast the best psychological profile: lowest Anxiety (~1.7), lowest Stress (~5.1), and highest Mood (~6.4) . Their platform of choice is Facebook .
- **Distinguishing Features:** Demographically and behaviorally distinct from the other two groups, they report the lowest online interaction counts (both positive and

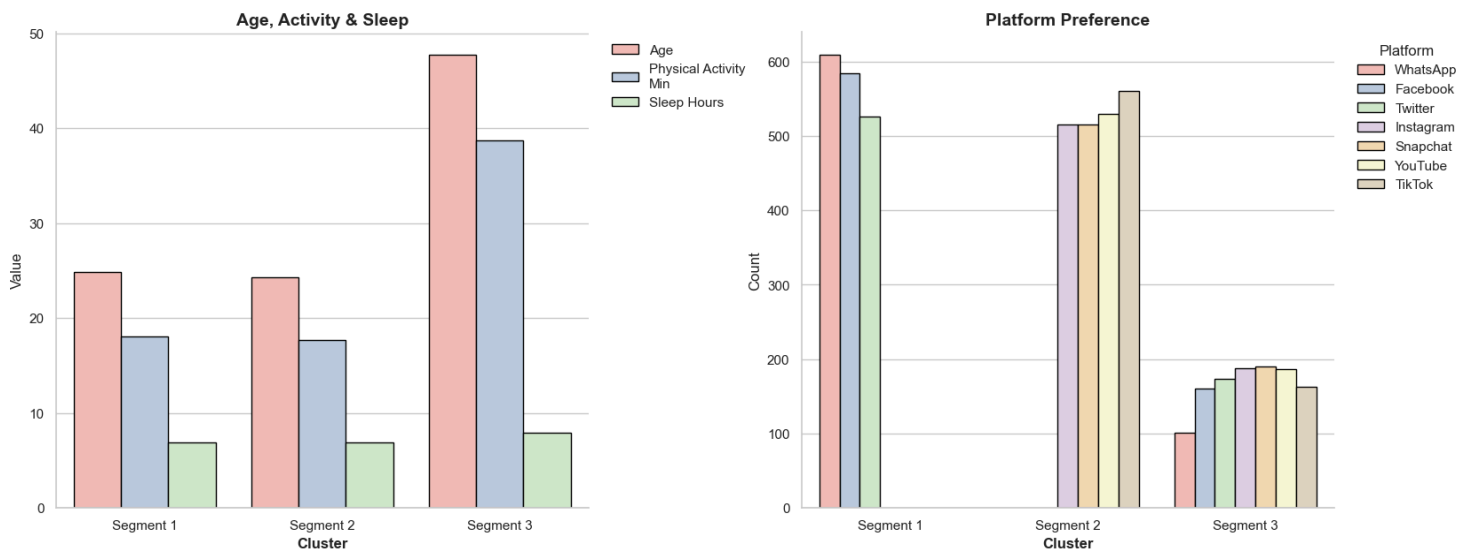
negative) . Crucially, this is the only cluster that contains a significant volume of users classified with a "Healthy" mental state .

- **Application Meaningfulness:** Highly meaningful. This cluster acts as the baseline "control group" for the dataset. It practically demonstrates that lower social media usage, combined with proactive offline habits (exercise and sleep), correlates strongly with stable mental health and higher overall mood.

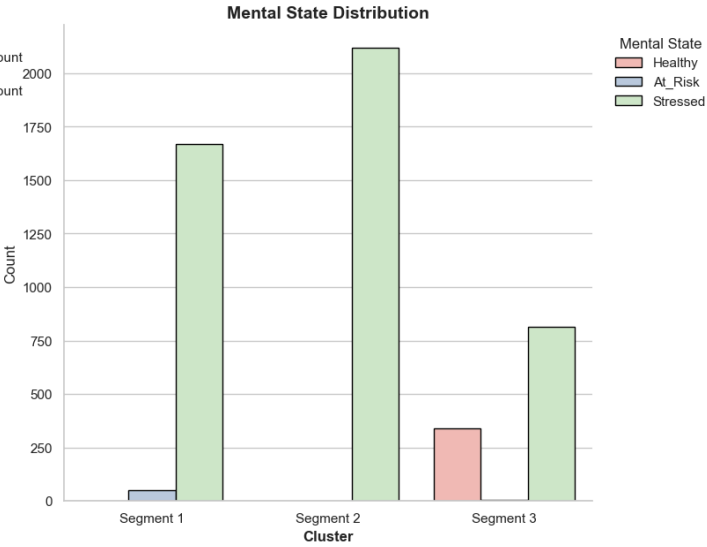
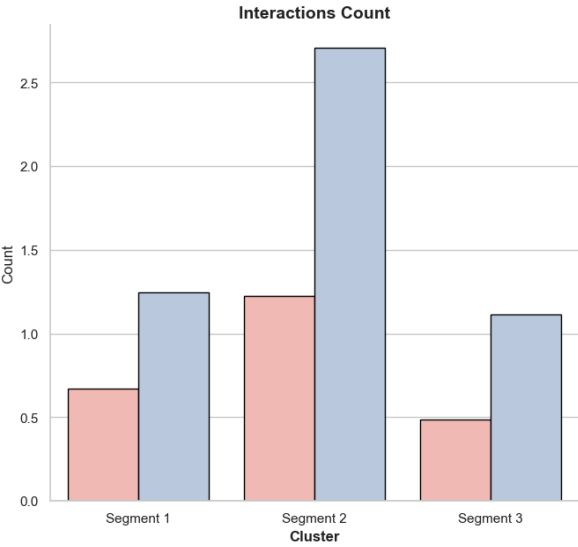
NMF (3 Components) - Optimal | GMM Profiling: Time & Psych (Optimal K=3)



NMF (3 Components) - Optimal | GMM Profiling: Health & Platform (Optimal K=3)



NMF (3 Components) - Optimal | GMM Profiling: Interactions & Mental State (Optimal K=3)



Conclusions and Discussion

Returning to our original project question, whether the choice of primary social media platform or user age correlates with anxiety or stress levels. The clustering analysis yielded invaluable insights into how modern populations segment themselves across digital environments. We learned that users do not merely vary linearly by screen hours. They cluster into distinct lifestyle profiles where specific platform mechanics fundamentally align with specific psychological states.

The results provide support for our original hypothesis and yielding three key conclusions:

- **Screen Time vs. Mental Strain:** There is a distinct, positive correlation between excessive daily screen time and severe psychological strain. Users within the "Heavy Social Media" segment demonstrate severe sleep deprivation and extreme anxiety scores.
- **Platform Architecture and Mental Strain:** Platform design matters just as much as raw screen time. Users favoring highly visual, algorithmically aggressive platforms like TikTok and Instagram drive the highest stress and anxiety. Conversely, those relying on functional communication apps like WhatsApp maintain significantly healthier physical activity levels and balanced sleep routines.
- **Age & Platform Preference:** Age proved to be a critical correlating factor. The "Healthy, Low-Digital" segment demonstrated that an older demographic (average age ~48) strongly correlates with a preference for legacy platforms like Facebook, healthier offline habits, lower overall screen time, and significantly lower mental strain.

However, several limitations must be noted. While our raw dataset is naturally non-negative, NMF's strict non-negativity constraint restricted our preprocessing options. It forced the exclusive use of Min-Max scaling to maintain a positive range and prevented the use of mean-centering or Standard Scaling. Furthermore, the GMM clustering algorithm, while powerful for its soft probabilistic assignments, remains highly sensitive to initialization parameters and can fall into local optimization traps.

For future directions, incorporating non-linear dimensionality reduction techniques could capture more delicate, complex behavioral structures. Additionally, integrating longitudinal time-series data would allow researchers to track cluster migration. It will help identify the exact behavioral inflection points where a stable "Utility Screen" user begins degrading into a high-risk "Heavy Social Media" state.

Metric	Utility User (cluster 1)	Heavy Social User (cluster 2)	Healthy Low Digital (cluster 3)
Average Age	25	24	48
Total Screen Time	380 minutes	390 minutes	210 minutes
Social Media Time	140 minutes	240 minutes	100 minutes
Stress / Anxiety	6.1 / 2.1	8.0 / 3.1	5.1 / 1.7
Primary Platforms	WhatsApp, Facebook, Twitter	TikTok, Youtube, Instagram, Snapchat	All platforms

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